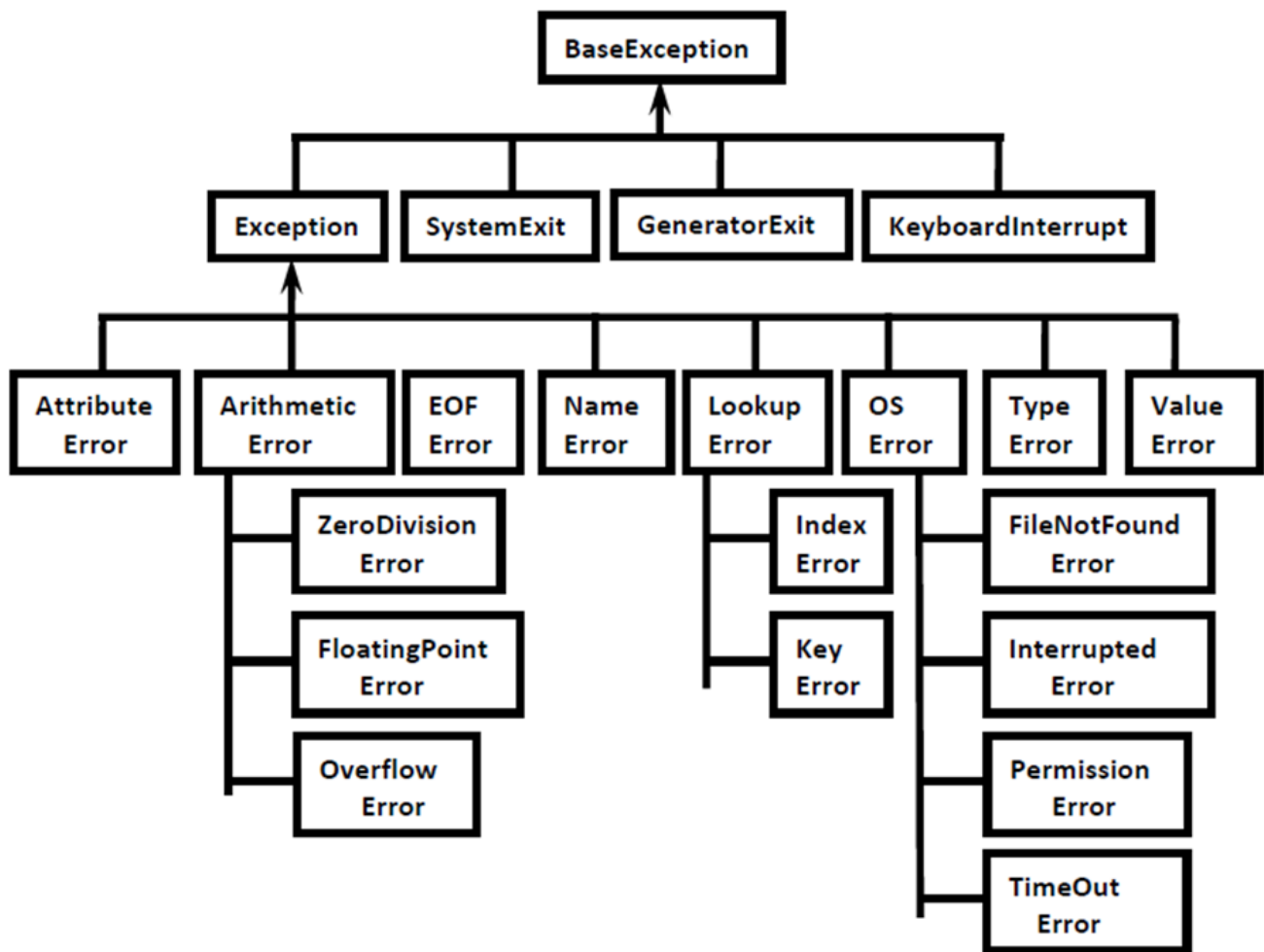


Error Handling



An unwanted or unexpected event that disturbs the normal flow of the program is called an exception. Whenever an exception occurs, immediately the program will terminate abnormally. In order to get our program executed normally, we need to handle those exceptions on high priority.

BaseException:

1. Exception
2. SystemExit ----- exit()
3. GeneratorExit ----- generator.close()
4. KeyboardInterrupt ----- ctrl+c

Exception:

1. Attribute Error :

AttributeError can be defined as an error that is raised when an attribute reference or assignment fails.

```
s = 'Python'
```

```
print(s.uper()) # its not uper(), its upper()
```

```
# O/P:--
```

```
Traceback (most recent call last):
```

```
File "C:\Users\Neeraj\OneDrive\Desktop\numpy_learning\p1.py", line 3, in <module>
```

```
    print(s.uper())
```

```
        ^^^^^
```

```
AttributeError: 'str' object has no attribute 'uper'. Did you mean: 'upper'?
```

2. Arithmetic Error: ZeroDivision, FloatingPoint, Overflow

ArithmeticError is simply an error that occurs during numeric calculations.

```
x=int(input('Enter any no :--- '))
```

```
y=int(input('Enter any no :--- '))
```

```
print(x/y)
```

```
O/P:--
```

```
PS E:\Python Core_Advance> py ex.py
```

```
Enter any no :--- 10
```

```
Enter any no :--- 0
```

```
Traceback (most recent call last):
```

```
File "E:\Python Core_Advance\ex.py", line 29, in <module>
```

```
    print(x/y)
```

```
ZeroDivisionError: division by zero
```

3. EOF Error----(End of file):-

EOF stands for "end of file," and this syntax error occurs when Python detects an unfinished statement or block of code. This can happen for many reasons, but the most likely cause is missing punctuation or an incorrectly indented block.

4. Name Error:-

In Python, a NameError: name 'x' is not defined error is raised when the program attempts to access or use a variable that has not been defined or assigned a value

```
print(x)
```

```
O/P:--
```

```
PS E:\Python Core_Advance> py ex.py
```

```
Traceback (most recent call last):
```

```
File "E:\Python Core_Advance\ex.py", line 37, in <module>
    print(x)
```

NameError: name 'x' is not defined

5. Lookup Error (Index, Key)

The LookupError exception in Python forms the base class for all exceptions that are raised when an index or a key is not found for a sequence or dictionary respectively.

```
# index error
```

```
l = [10, 20, 30]
print(l[5])
```

```
# o/p :
```

```
File "C:\Users\Neeraj\OneDrive\Desktop\p1.py", line 2, in <module>
    print(l[5])
    ~^^^
```

IndexError: list index out of range

```
d = {"name": "Neeraj"}
print(d["age"])
```

```
File "C:\Users\Neeraj\OneDrive\Desktop\numpy_learning\p1.py", line 5, in <module>
    print(d["age"])
    ~^^^^^^^
```

KeyError: 'age'

6. OS Error

```
f = open('neeraj.txt')
data = f.read()
print(data)
f.close()
```

O/P:--

PS E:\Python Core_Advance> py file.py

Traceback (most recent call last):

```
File "E:\Python Core_Advance\file.py", line 97, in <module>
    f = open('neeraj.txt')
```

FileNotFoundError: [Errno 2] No such file or directory: 'neeraj.txt'

7. Type Error

TypeError is raised whenever an operation is performed on an incorrect/unsupported object type. For example, using the + (addition) operator on a string and an integer value will raise a TypeError.

```
x=input('Enter any str :--- ')
y=input('Enter any str :--- ')
print(x*y)

O/P:--
PS E:\Python Core_Advance> py ex.py
Enter any str :--- neeraj
Enter any str :--- kumar
Traceback (most recent call last):
  File "E:\Python Core_Advance\ex.py", line 29, in <module>
    print(x*y)
TypeError: can't multiply sequence by non-int of type 'str'
```

```
# -----
```

```
x = 10
y = '20'

print(x+y)
```

```
O/P:-- Traceback (most recent call last):
  File "C:\Users\Neeraj\OneDrive\Desktop\numpy_learning\p1.py", line 4, in <module>
    print(x+y)
      ~^~
TypeError: unsupported operand type(s) for +: 'int' and 'str'
```

8. Value Error:-

```
x=int(input('Enter any no :---'))
print(x)

O/P--
PS E:\Python Core_Advance> py ex.py
Enter any no :--- Neeraj
Traceback (most recent call last):
  File "E:\Python Core_Advance\ex.py", line 27, in <module>
    x=int(input('Enter any no :--- '))
ValueError: invalid literal for int() with base 10: 'Neeraj'
```

```
# -----
```

```
s = 'Python'

print(int(s))
```

```
# O/P:--
```

```
PS C:\Users\Neeraj\OneDrive\Desktop> py p1.py
```

```
Traceback (most recent call last):
```

```
File "C:\Users\Neeraj\OneDrive\Desktop\numpy_learning\p1.py", line 3, in <module>
    print(int(s))
          ~~~^^^
```

```
ValueError: invalid literal for int() with base 10: 'Python'
```